CAPITAL BUDGETING

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Learning Outcomes

- Understand the purpose and importance of capital budgeting in business decision-making.
- Identify and analyze cash flows associated with long-term investment projects.
- Apply key evaluation methods like NPV, IRR, and Payback Period to assess project viability.
- Incorporate the time value of money in investment appraisals.
- Evaluate risks and uncertainties impacting capital investment decisions.
- Make informed recommendations for selecting profitable and strategic projects.

Capital Budgeting

Capital budgeting is the process by which companies evaluate and decide on long-term investments and projects. These investments usually involve significant amounts of money, such as:

- Buying new machinery or equipment
- Building a new factory
- Launching a new product line
- Expanding operations

The goal is to determine whether a project or investment is worth pursuing based on its potential to generate profits over time.

Capital Budgeting

Common examples of capital budgeting decisions:

- Building a new factory
- Buying new machinery
- Launching a new product
- Acquiring another company
- Expanding into a new market

Objectives of Capital Budgeting

1. Maximize shareholder wealth
2. Evaluate and select the best investment opportunities
3. Ensure efficient allocation of capital resources
4. Minimize risk by assessing future uncertainties

Steps in the Capital Budgeting Process

1.Project Identification

1.Project Evaluation

1.Project Selection

1.Project Implementation

1.Performance Review

Steps in the Capital Budgeting Process

1. Project Identification
   - Identify potential investment opportunities (e.g., building a new plant, purchasing machinery).
2. Project Evaluation
   - Forecast future cash flows of each project.
   - Evaluate risk and strategic alignment.
3. Project Selection
   - Apply quantitative techniques to compare projects.
   - Select the project(s) that add the most value.
4. Project Implementation
   - Allocate resources and start executing the project.
5. Performance Review
   - Monitor outcomes and compare with projections.
   - Learn from discrepancies.

Capital Budgeting Techniques

1. Net Present Value (NPV)
2. Internal Rate of Return (IRR)
3. Payback Period
4. Discounted Payback Period

Net Present Value (NPV)

Net Present Value (NPV) is the difference between the present value of cash inflows and the present value of cash outflows over a period of time. It is used to determine the profitability of an investment or project.

NPV Formula :

$$\mathrm{NPV} = \sum_{t=1}^{n} \frac{CF_t}{(1+r)^t} - c_0$$

WhereType equation here.

- CF = Cash inflow at time t
- n = Number of periods
- C0 = Initial investment (cash outflow at time 0)
- r= Discount rate (cost of capital)
- t = Time period (1, 2, 3…n)

Key Concepts in NPV

1. Time Value of Money (TVM)

- A rupee today is worth more than a rupee tomorrow.
- NPV discounts future cash flows to reflect this.

2. Discount Rate

- Usually, the company's cost of capital or required rate of return.
- Reflects the opportunity cost and risk of the investment.

3. Cash Flows

- Only relevant (incremental) cash flows are considered — not accounting profits.

NPV Decision Rule

- NPV > 0 → Accept the project (adds value to the firm)
- NPV < 0 → Reject the project (destroys value)
- NPV = 0 → Indifferent (no gain, no loss)

Example of NPV Calculation

Project: A company is considering investing ₹ 10,000 in a project expected to generate the following cash inflows over 4 years. The company's required rate of return (discount rate) is 10%.

Year	Cash Inflow (CF)
1	₹3,000
2	₹4,000
3	₹4,000
4	₹2,000

Example of NPV Calculation

Answer: Step 1: Write down the formula $\mathrm{NPV} = \sum_{t=1}^{4} \frac{CF}{(1+0.10)^t} - 10{,}000$

Step 2: Calculate present value (PV) of each cash flow

Year	CF	Discount Factor $\frac{1}{(1+0.10)^t}$	Present Value (PV) = CF × Discount Factor
1	3,000	0.9091	3,000 × 0.9091 = 2,727.30
2	4,000	0.8264	4,000 × 0.8264 = 3,305.60
3	4,000	0.7513	4,000 × 0.7513 = 3,005.20
4	2,000	0.6830	2,000 × 0.6830 = 1,366.00

Example of NPV Calculation

- Step 3: Sum the present values of cash inflows

2,727.30+3,305.60+3,005.20+1,366.00=10,404.10

- Step 4: Subtract initial investment

NPV = 10,404.10 -10,000=404.10

- Conclusion:

Since NPV = ₹ 404.10 > 0, the project is expected to add value and should be accepted.

Question

Question 1:

- Initial Investment: ₹ 5,000
- Cash Inflows:
  Year 1: ₹ 2,000
  Year 2: ₹ 2,500
  Year 3: ₹ 2,500
- Discount Rate: 10%

Question 2:

- Initial Investment: ₹ 6,000
- Cash Inflows:
  Year 1: ₹ 1,500
  Year 2: ₹ 1,800
  Year 3: ₹ 2,000
- Discount Rate: 12%

Advantages of NPV

- Considers the time value of money
- Accounts for all cash flows
- Uses objective decision rule (maximizing shareholder value)
- Adjusts for risk via discount rate

Disadvantages of NPV

- Requires accurate estimation of future cash flows
- Sensitive to the discount rate used
- Not easily comparable across projects of different sizes
- Doesn't always work well with mutually exclusive or non-conventional cash flows

Best Practices When Using NPV

- Use realistic cash flow projections.
- Select an appropriate discount rate (usually the cost of capital).
- Use scenario/sensitivity analysis for uncertain variables.
- Combine with other methods like IRR or Payback for better decision-making.

When to Use NPV

- Evaluating large capital investments
- Comparing multiple long-term projects
- Making acquisition decisions
- Budgeting for R&D or infrastructure

Questions

How do you decide which projects to prioritize when capital is limited?

Answer

When capital is limited, projects should be prioritized based on which ones offer the highest Net Present Value (NPV) or profitability index—meaning the most value created per dollar invested. This ensures the company maximizes returns on scarce resources. Besides financial metrics, it’s important to consider strategic alignment with business goals, risk levels, and resource availability, like manpower or technology. Timing of cash flows also matters; projects with quicker returns may be preferred to improve liquidity. Combining these factors helps make balanced, value-driven investment decisions under capital constraints.

IRR (Internal Rate of Return)

- The Internal Rate of Return (IRR) is the discount rate that makes the Net Present Value (NPV) of a project zero.
- In simpler terms, it's the rate of return at which the investment breaks even in terms of present value.

How to Calculate IRR?

- The IRR is the value of r (discount rate) that satisfies this equation :

Basic formula: $\mathrm{NP}V == \sum_{t=1}^{n} \frac{CF_t}{(1+r)^t} - c_0$

Key Characteristics of IRR

- IRR is expressed as a percentage (%).
- It represents the annualized rate of return expected from a project.
- It is the breakeven discount rate — where investment equals return.

IRR vs Cost of Capital (r)	Decision
IRR > r	Accept the project
IRR < r	Reject the project
IRR = r	Indifferent

How to Calculate IRR

Question:

- Invest ₹ 8,000 now. Receive ₹ 3,000 each year for 3 years. Find IRR.

Answer:

- Step 1: Use IRR formula:

$$= -8{,}000 + \frac{3000}{(1+r)} + \frac{3000}{(1+r)^1} + \frac{3000}{(1+r)^3}$$

- Step 2: Trial with 6%:

$$\text{NPV} = -8{,}000 + \frac{3000}{1.06} + \frac{3000}{1.1236} + \frac{3000}{1.1910}$$

NPV > 0 → IRR > 6%

Example: IRR Calculation

- Step 3: Trial with 8%: NPV = -8,000 $\frac{3000}{1.08} + \frac{3000}{1.1664} + \frac{3000}{1.2597}$

Step 4: Interpolate: IRR = 6%$\left(\frac{19.45}{19.45+269.77}\right) \times (8\% - 6\%) = 6\% + \frac{19.45}{289.22} \times 2\%$

=6%+0.134%=6.13%

Advantages of IRR

- Easy to understand: Shows return as a simple percentage.
- Accounts for time value of money.
- Considers all cash flows.
- Useful for comparing projects of different sizes.

Disadvantages of IRR

1. Multiple IRRs:

- Occurs if the project has non-conventional cash flows (sign changes more than once).
- May lead to confusion or a wrong decision.

2. Assumes reinvestment at IRR:

- Unrealistic assumption compared to NPV (which assumes reinvestment at the cost of capital).

Disadvantages of IRR

3. Ignores the scale of the project:

- A small project with a high IRR might create less value than a larger project with a lower IRR.

4. Not suitable for mutually exclusive projects:

- May lead to wrong decisions compared to NPV.

When to Use IRR

- You want a quick estimate of return as a percentage.
- Comparing projects of similar size and duration.
- Communicating performance to non-financial stakeholders.
- Complementing NPV analysis (not replacing it)

NPV vs IRR

Feature	NPV	IRR
Type	Absolute value (₹)	Percentage rate (%)
Decision Rule	Accept if NPV > 0	Accept if IRR > cost of capital
Reinvestment Assumption	Reinvest at cost of capital	Reinvest at IRR
Reliability	More reliable for unique decisions	Can be misleading in special cases
Multiple Rates Issue	No	Possible with non-conventional cash flows

Question

"*How can the IRR method lead to incorrect investment decisions when evaluating mutually exclusive projects?*"

Answer

IRR can be misleading for mutually exclusive projects because it shows the rate of return, not the actual value created. A project with a higher IRR might have a lower NPV, meaning it adds less value. Also, IRR assumes reinvestment at the IRR itself, which is unrealistic. So, NPV is more reliable for decision-making

Questions

1. What happens if a project has multiple IRRs?

2. Why is NPV preferred over IRR in capital budgeting? Can a project have no IRR?

3. What does it mean if IRR = cost of capital?

4. How do timing differences in cash flows affect IRR and NPV?

5. Can IRR and NPV give conflicting decisions? When and why?

Payback Period

- The Payback Period is the amount of time it takes for an investment to recover its initial cost from the cash inflows it generates.

Payback Period Formula:

- If cash inflows are equal each year:

$$\text{Payback Period} = \frac{Initial\ Investment}{Annual\ Cash\ Inflow}$$

If cash inflows are unequal:

- You must cumulatively add the cash inflows each year until the total equals the initial investment.

Payback Period Decision Rule

- If Payback Period ≤ Target Payback Period → Accept the project
- If Payback Period > Target → Reject the project

Example: If a company's policy is to accept only projects with a payback period under 3 years, a project with a payback of 2.5 years would be accepted.

Example 1: Equal Cash Flows

- Initial Investment = 40,000 ₹
- Annual Cash Inflow = 10,000 ₹
- Payback Period $= \frac{40,000}{10,000}$ =4 years

Advantages of Payback Period

1. Simple to calculate and understand.
2. Emphasizes liquidity — useful for firms with cash flow concerns.
3. Good for quick, preliminary screening of projects.
4. Reduces risk by focusing on how quickly the investment is recovered.

Disadvantages of Payback Period

Weakness	Explanation
Ignores Time Value of Money	₹1 today ≠ ₹1 in future — not considered in basic payback.
Ignores Cash Flows After Payback	Doesn’t consider long-term profitability.
Arbitrary Benchmark	Target payback period may not reflect actual risk or opportunity cost.
Not Ideal for Strategic or Long-Term Projects	Can lead to rejecting valuable long-term investments.

Summary

Feature	Payback Period
Focus	Liquidity / Risk
Easy to Use?	Yes
Considers TVM?	No (Yes if discounted)
Long-Term Focus?	No
Decision Rule	Accept if payback ≤ cutoff

Payback Period vs Other Methods

Method	Time Value of Money	Considers All Cash Flows	Output	Best Used For
Payback Period	No	No	Years	Quick liquidity analysis
Discounted Payback	Yes	No	Years	Risk-sensitive liquidity analysis
NPV	Yes	Yes	₹ Value	Profit-maximizing decisions
IRR	Yes	Yes	% Return	Comparing profitability

Discounted Payback Period

- The Discounted Payback Period is the amount of time it takes for the present value of an investment's cash inflows to recover the initial investment.
- Unlike the regular payback period, it discounts each cash flow using a required rate of return or cost of capital before summing.

How to Calculate Discounted Payback Period:

1. Determine the discount rate (usually cost of capital).
2. Calculate the present value (PV) of each cash inflow:

Discounted Payback Period

Basic Formula: $$\mathrm{PV} = \frac{CF}{(1+r)^t}$$

Where: cash flow at time t

- r = discount rate
- t = time period

3. Cumulatively add discounted cash inflows until the sum equals or exceeds the initial investment.
4. The discounted payback period is the time it takes for this to happen, including fractions of the year.

Example

- Initial Investment: ₹ 10,000
- Cash Inflows:
  - Year 1: ₹ 3,000
  - Year 2: ₹ 4,000
  - Year 3: ₹ 4,000
- Discount Rate: 10%
- Step 1: Calculate discounted cash inflows

Year	Cash Flow	Discount Factor (10%)	Discounted Cash Flow (DCF)
1	3,000	1 / (1.10)^1 = 0.9091	3,000 × 0.9091 = 2,727.30
2	4,000	1 / (1.10)^2 = 0.8264	4,000 × 0.8264 = 3,305.60
3	4,000	1 / (1.10)^3 = 0.7513	4,000 × 0.7513 = 3,005.20

Example

Step 2: Calculate cumulative discounted cash inflows

Year	Discounted Cash Flow	Cumulative DCF
1	2,727.30	2,727.30
2	3,305.60	2,727.30 + 3,305.60 = 6,032.90
3	3,005.20	6,032.90 + 3,005.20 = 9,038.10

Example

Step 3: Find when cumulative DCF recovers the initial investment (₹ 10,000)

- After Year 3, cumulative discounted cash inflows = ₹ 9,038.10 (still less than ₹ 10,000)
- So, the investment is not fully recovered by Year 3.

Step 4: Since Year 4 inflow is not given, assume another ₹ 4,000 for Year 4

- Year 4 discounted cash flow: $\frac{4{,}000}{(1.10)^4} = \frac{4{,}000}{1.4641} = 2{,}731.0$
- Add to cumulative: $9{,}038.10 + 2{,}731.0 = 11{,}769.10 > 10{,}000$

Example

Step 5: Calculate the fractional year to recover the remaining amount:

- Remaining amount at end of Year 3 = ₹ 10,000 - 9,038.10 = ₹ 961.90

- Fraction of Year 4 needed $= \frac{961.90}{2,731.0}$ =0.35 years

Advantages of Discounted Payback Period

- Considers the time value of money.
- Helps evaluate investment liquidity while adjusting for risk and cost of capital.
- Useful in risk management, especially where early recovery of investment is critical.

Disadvantages

1. Ignores cash flows that occur after the payback period (just like traditional payback).
2. More complex to calculate than simple payback.
3. Can still reject projects with positive NPV if payback takes too long.
4. Not a complete measure of profitability.

When to Use Discounted Payback Period

- When time value of money matters but you still want a liquidity measure.
- For projects where early cash recovery is important.
- In combination with other metrics like NPV and IRR.

Discounted Payback vs Payback Period

Feature	Payback Period	Discounted Payback Period
Time Value of Money	No	Yes
Complexity	Simple	More complex
Decision Focus	Liquidity	Liquidity + TVM
Considers Full Cash Flows	No	No
Practical Use	Quick screening	More accurate screening

Summary

Step	Description
1. Discount cash flows	Calculate PV of each cash inflow
2. Cumulative sum	Add discounted cash flows cumulatively
3. Determine payback	Identify when cumulative PV equals initial cost
4. Calculate fraction	Find fraction of the final year needed

Payback Period vs Other Methods

Method	Time Value of Money	Considers All Cash Flows	Output	Best Used For
Payback Period	No	No	Years	Quick liquidity analysis
Discounted Payback	Yes	No	Years	Risk-sensitive liquidity analysis
NPV	Yes	Yes	₹ Value	Profit-maximizing decisions
IRR	Yes	Yes	% Return	Comparing profitability

Questions

Question 1:

If a project has a shorter payback period but a lower NPV compared to another project,

which should a company choose? Why?

Answer

The company should generally choose the project with the **higher NPV**, even if it has a longer payback period. NPV measures the total value added by the project, reflecting profitability and wealth creation. A shorter payback period focuses only on liquidity and risk recovery but ignores the overall profitability and cash flows beyond the payback point. However, if the company faces liquidity constraints or higher risk aversion, the shorter payback might be preferred.

Questions

Question 2:

Why might the discounted payback period still be insufficient for making investment decisions despite considering the time value of money?

Answer

Though DPP accounts for the time value of money, it **ignores cash flows beyond the payback period**, which can be significant. This limitation can lead to rejecting projects that are profitable in the long run but recover the investment slowly. It also doesn't measure overall profitability or compare the scale of projects, so relying solely on DPP can result in suboptimal decisions.